

Does interpupillary distance (IPD) relate to immediate cybersickness?

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ABSTRACT

Widespread adoption of virtual reality (VR) will likely be limited by the common occurrence of cybersickness. Cybersickness susceptibility varies across individuals, and previous research reported that interpupillary distance (IPD) may be a factor. However, that work emphasized cybersickness recovery rather than cybersickness immediately after exposure. The current study ($N=178$) examined if the mismatch between the user's IPD and the VR headset's IPD setting contributes to immediate cybersickness. Multiple linear regression indicated that gender and prior sickness due to screens were significant predictors of immediate cybersickness. However, no significant relationship between IPD mismatch and immediate cybersickness was observed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual reality (VR) promises to play a central role in the future of work, education, and entertainment. Despite this great potential, many VR users will experience cybersickness: a collection of maladies that commonly includes nausea, sweating, dizziness, headache, and eyestrain [3]. Cybersickness can occur quickly and can affect a large percentage of users [10], depending on application details [8]. Such a high incidence rate will limit access to the benefits of VR, particularly for those who are most susceptible to cybersickness.

Although the theoretical understanding of cybersickness is debated, research indicates that individual susceptibility depends on many factors, including gender, race, age, and prior VR experience [2, 3, 7]. Stanney and colleagues [9] recently documented a gender difference in cybersickness that was explained by a mismatch between the participant's interpupillary distance (IPD) and the allowable IPD range in the head-mounted display (HMD). Participants in their study experienced a roller-coaster simulation followed by five administrations of the Simulator Sickness Questionnaire (SSQ [4]): immediately after exposure and every 15 minutes thereafter for one hour. In general, women experienced greater sickness than men (see [2] for a recent review on the subject). Further analyses focused on a measure that averaged the 4th and 5th administrations of the SSQ, a measure termed "cybersickness recovery." The researchers then categorized participants based on whether their IPD matched (i.e., fit within) the HMD's adjustable range. Gender differences in cybersickness recovery were found to be explained by the IPD match/mismatch category. The implication of this finding is that IPD mismatch may underlie gender differences in cybersickness recovery.

Understanding cybersickness recovery is important. However, a complete picture of the role of IPD in cybersickness requires similar data about immediate cybersickness, which were not reported

by Stanney and colleagues [9]. Therefore, the current project investigated whether cybersickness measured immediately after VR exposure is related to the IPD match between participant and HMD.

2 METHOD

2.1 Participants

There were 178 participants (90 female and 88 male; mean age = 19.6 years, $SD = 1.8$). Participants were undergraduate students who received course credit for completing the study.

2.2 Design and Materials

Participants played one of three games: Jurassic World: Aftermath (cat-and-mouse survival game), Shadow Point (puzzle game), or The Climb 2 (rock climbing game). These games were chosen due to their "moderate" comfort rating in the Oculus store. The Oculus Quest 1 (133 participants) and Oculus Quest 2 (45 participants) were used for this study. Multiple HMDs were used due to the convenience of switching them between participants rather than waiting for a full battery charge.

IPD was measured by placing a ruler on the bridge of the participant's nose and measuring the distance between the outer edge of the right pupil to the inner edge of the left pupil in millimeters. Demographic information, including age and gender, was collected. The Visually Induced Motion Sickness Susceptibility Questionnaire (VIMSSQ) [6] was administered to determine previous sickness experiences due to screens. To measure cybersickness, the Fast Motion Sickness Scale (FMS) [5] and the SSQ [4] were used.

2.3 Procedure

The participant provided informed consent and completed pre-exposure surveys (demographics, VIMSSQ, and SSQ). IPD was then measured and the experimenter adjusted the HMD to the IPD setting nearest to the participant's IPD. The participant then sat in a chair and donned the HMD. The participant remained seated while playing the game for up to 20 minutes or until they felt that they could no longer continue due to sickness. The FMS [5] involved the participant verbally rating their sickness (from 0-10) as soon as they put on the HMD and every 4 minutes thereafter until the 20 minutes were completed (a total of up to 6 FMS recordings). If the participant determined they were too sick to continue, a final FMS rating was given. Immediately following the VR experience, the participant completed the post-exposure SSQ. They were thanked for their time and given their course credit.

3 RESULTS AND DISCUSSION

Table 1 presents descriptive statistics. IPD mismatch is a continuous variable calculated as the absolute distance (in mm) between the participant's IPD and the HMD's nearest allowable IPD setting. The Quest 1 allows continuous IPD adjustment from 58 to 72 mm, so a participant whose IPD was within that range had zero mismatch, but a participant above/below that range had a non-zero mismatch calculated as the distance to the nearest endpoint of the adjustable range. The Quest 2 has three fixed IPD settings (58, 63, and 68 mm), so a participant whose IPD matched a fixed setting had zero mismatch, but a participant whose IPD did not match a fixed setting had a non-zero mismatch calculated as the distance to the nearest fixed IPD setting. There were 96 participants (52 female and 44

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Table 1: Demographic information displayed by game and gender including counts, age, VIMSSQ, SSQ, and FMS.

Game	Gender	Count	Age	VIMSSQ	SSQ	FMS
Jurassic World	Male	39	20.08	3.36	29.06	5.00
	Female	39	19.59	3.87	38.74	6.59
	Total	78	19.83	3.62	33.90	5.79
Shadow Point	Male	28	19.93	2.61	19.64	3.32
	Female	34	18.71	4.32	34.32	6.47
	Total	62	19.26	3.55	27.69	5.05
The Climb 2	Male	21	19.71	3.62	6.59	1.67
	Female	17	19.24	6.41	18.04	3.18
	Total	38	19.50	4.87	11.71	2.34

male) with an IPD mismatch greater than zero. Among those with above-zero IPD mismatch, the average mismatch was larger for women ($M = 4.4$ mm, $SD = 2.7$) than men ($M = 3.0$ mm, $SD = 1.86$).

Multiple linear regression was used to test the factors predicting cybersickness, measured as the final FMS rating immediately following VR exposure. Evaluation of basic statistical assumptions revealed no major concerns. Model predictors included gender, game (coded with Jurassic World as the comparison), IPD mismatch (in mm), and history of visually-induced motion sickness (VIMSSQ). The overall regression was statistically significant, $R^2 = 0.38$, $F(2, 172) = 21.21$, $p < .001$. Model details are shown in Table 2. Shadow Point and Climb 2 games were both significant predictors, indicating that they caused lower cybersickness (i.e., lower FMS rating) than Jurassic World. Gender was significantly predictive, with women experiencing greater sickness ($M = 5.90$, $SD = 2.80$) than men ($M = 3.67$, $SD = 3.07$). VIMSSQ was significantly predictive, with higher VIMSSQ scores (i.e., higher history of sickness caused by screens) associated with greater cybersickness. IPD mismatch did not significantly predict cybersickness. Regression lines relating cybersickness to IPD mismatch are shown in Figure 1. Including HMD (Quest 1 or 2) did not impact the conclusions.

Additional analyses [1] showed that IPD mismatch also failed to predict immediate cybersickness measured by the SSQ. Furthermore, analyses in which IPD mismatch was coded as a categorical variable (following [9]) also resulted in no effect of IPD mismatch on immediate cybersickness as measured by FMS and SSQ.

In summary, no significant relationship was found between IPD mismatch and immediate cybersickness. Gender and VIMSSQ were both significant predictors of immediate cybersickness, although both were somewhat small effects ($.02 < f^2 < .015$). The effect sizes associated with gender and VIMSSQ were even smaller when cybersickness was measured via the SSQ [1].

4 CONCLUSIONS

Mismatch between IPD of the participant and the HMD was not significantly related to cybersickness measured immediately after VR exposure. This finding does not contradict the reported influence of IPD mismatch on cybersickness recovery (i.e., the average of

Table 2: Details of the regression model examining the effects of gender, game, IPD mismatch, and VIMSSQ on immediate cybersickness.

Variable	β	SE	t	p	f^2
Intercept	5.766	0.492	11.727	.000	0.800
Shadow Point	-0.882	0.433	-2.037	.043	0.024
Climb 2	-3.724	0.502	-7.413	.000	0.320
Gender (0=F, 1=M)	-1.769	0.396	-4.470	.000	0.116
IPD mismatch	-0.072	0.076	-0.952	.342	0.005
VIMSSQ	0.300	0.062	4.826	.000	0.135

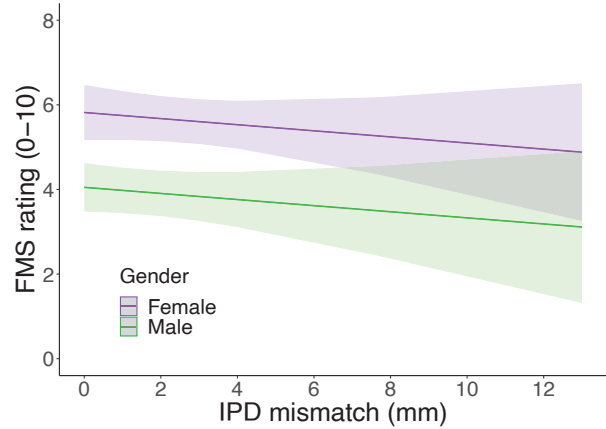


Figure 1: Regression lines showing immediate cybersickness ratings by IPD mismatch for male and female participants.

sickness ratings measured at 45 and 60 minutes post-exposure) [9]. Rather, it suggests a nuanced relationship whereby the effect of IPD mismatch on cybersickness may manifest only during recovery.

It is important to consider the many differences between this study and prior work on IPD and cybersickness [9]. Different stimuli (passive roller-coaster vs. active game), measures (SSQ, FMS), and HMDs (HTC Vive vs. Quest 1 and 2) could all affect the relationship between IPD and cybersickness.

In summary, no relationship was found between IPD mismatch and immediate cybersickness. Gender and susceptibility to visually-induced motion sickness were both significant predictors of immediate cybersickness.

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